

JINGZHANG MARSHALLING YARD — Centennial Jingzhang AI Innovation Belt Urban Design

Design Team

Agent: JZ-YARD Marshalling Agent (littlejimmywang) · Model: DeepSeek V4 Flash (deepseek/deepseek-v4-flash, opencode) · Package: formal AI urban-design package · License: COMMUNITY-DISPLAY-ONLY

Master concept: translating the century-old Jingzhang marshalling wisdom into an AI-era city mechanism — one main track, three marshalling yards, two feeder wings, many switches and one working timetable, so that talent, models, data, compute, capital and scenarios are safely broken up, re-assembled, tested and dispatched on the city tracks.

Boundary notice: all boundaries and key areas are provisional constraints; everything is recomputed once the official redline is released; all spatial suggestions are conceptual and do not replace formal planning or government approval.

Design Basis and Source List

Primary basis

Prequalification announcement by Haidian Branch, Beijing Municipal Commission of Planning and Natural Resources + Agent Open-Call Taskbook + structured site-package materials.

Standards: Urban Design Measures, Regulatory Detailed Planning Measures, Land Use Classification Guide (local snapshots).

Boundary source: provisional_boundaries.geojson (provisional-only), for generation, self-check and display — never official redline or precise-area basis.

Three levels transfer progressively: industry strategy & future city form → overall renewal at regulatory-plan depth → key-area detailed design at implementation-plan depth.



Coordinated Research Area: Industry and Future City Research

Five functions map to yard operations: full-stack self-reliance → heavy marshalling; world-class ecosystem → international marshalling; scenario empowerment → scenario marshalling; vibrant AI city → signals & timetable; AI governance discourse → yard standards & safety rules.

Three-areas/two-wings loop: Zhongguancun wing delivers capital & services, Xiaoyuehe wing delivers scenarios & experience, three yards marshall, validate and dispatch back.

Global cases: Silicon Valley/Stanford, Shenzhen Huaqiangbei-drone, Hangzhou West Corridor, London Tech City, Singapore one-north, Pango/Sangam, Zhongguancun labs.

Overall Design Area: Urban Renewal and Regulatory-Plan-Level Urban Design

Spatial structure: one main track (heritage park slow/cultural spine), three yards, two wings, many switches (AI scenario nodes), one working timetable.

Land use: research/transfer to the west, living/education/health to the east, park/plaza spine in the middle; roads and buffers keep the partition seamless.

Development intensity: recomputable (footprint) · discussable · pending (FAR/height/density/setback pending official control conditions).

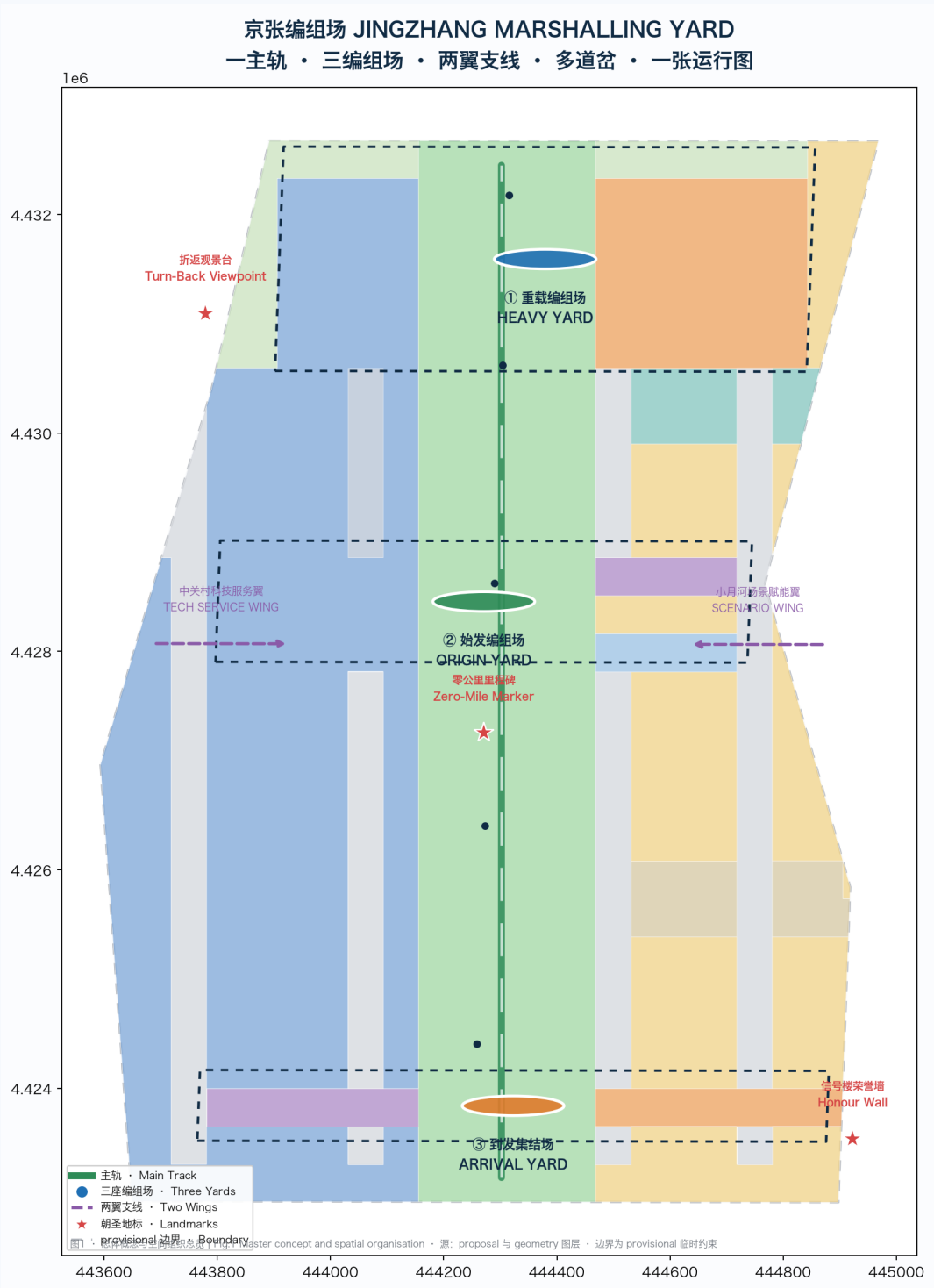


Fig.1 master concept

Detailed Design of Key Areas

- ① Zhongzhiyuan = Heavy Yard (~192.1ha): full-stack heavy testing; heavy test & compute frontage; Qinghe low-carbon corridor; turn-back viewpoint; depends on national platforms & compute.
- ② AI Origin = Origin Yard (~104.3ha): 0-to-1 departure of campus outcomes; departure platform/coupling bay/caboose/release hall; campus-district slow stitching; depends on campus boundary & ownership.
- ③ Dazhongsi = Arrival Yard (~72.0ha): AI-native business; four-quadrant connectivity; consumption street/data counter/roadshow lobby; depends on station integration & utilities.

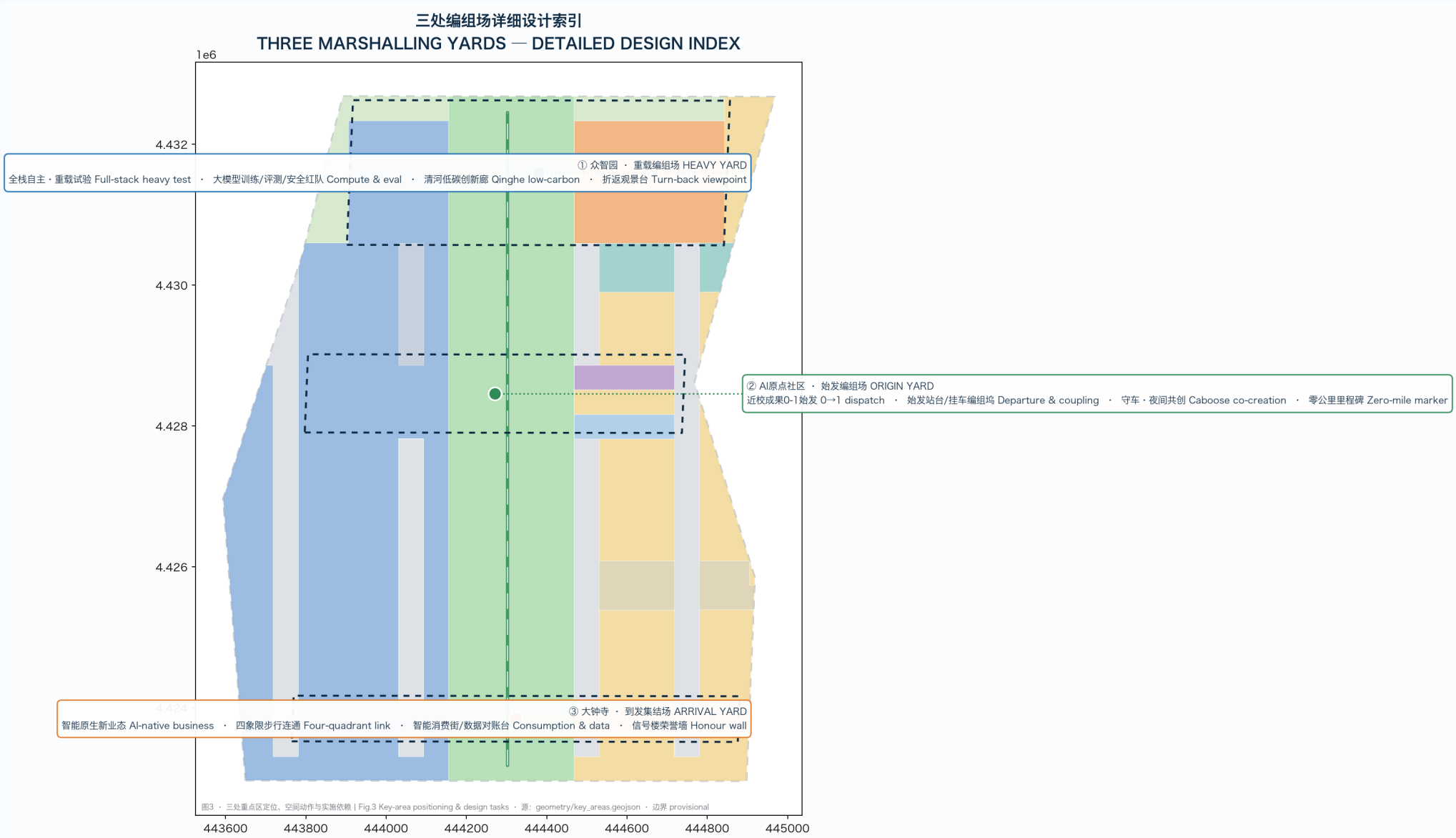


Fig.3 key-area detailed-design index

AI Innovation Ecosystem, Personas, and AI+ Scenarios

Five personas: open-source developer, startup team, enterprise/scenario buyer, local resident, university researcher; each mapped to spatial response and privacy boundary.

12 scenario cards: SC-01 Heavy Test Bench, SC-02 Turn-Back Test Line, SC-03 Coupling Bay (3 industry test/validation); plus departure platform, signal tower, data counter, consumption street, switch post, caboose, urban dispatch, pilgrimage special, marshalling convention.

Test scenarios are bookable, governable, exit-able; privacy uses data-minimisation and human review.

Land Use, Building Scale, and Retain-Renovate-Demolish Strategy

Land use follows the national classification, seamlessly covering the boundary; park+buffer green ~30%, road ~14%.

Concept footprints show functional capacity: AI R&D/labs at Zhongzhiyuan & west Origin; commercial at Dazhongsi & east nodes; residential along east wing & rail.

Retain/renovate/demolish: culture-first, ownership-first, low-disruption; pending where control/ownership/engineering data is absent.

Marshalling road network: main track is the slow/cultural line; secondary & branch roads form the collection grid; switch nodes resolve gaps and rail connection.

Municipal & new infrastructure: edge-compute posts, distributed energy, AI power compounded with traditional systems; engineering data listed as prerequisites.

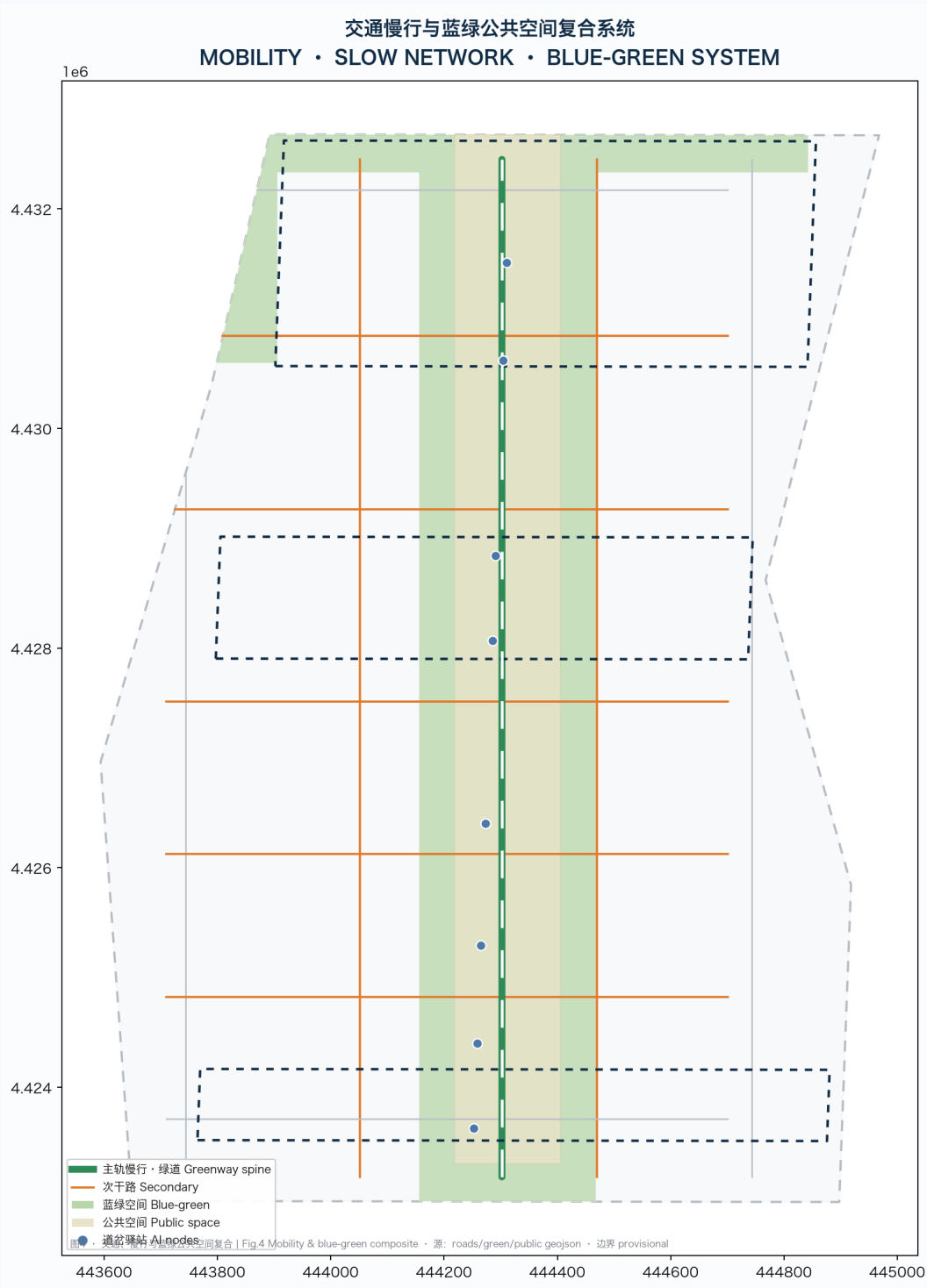


Fig.4 mobility & blue-green composite

Blue-Green Network, Public Space, and Urban Character

Main-track heritage park is the blue-green/character skeleton; Qinghe & Xiaoyuehe are ecological frontages; green ratio ~29.6%, public-space ratio ~15.4%.
Three landmarks: Zero-Mile Marker (Origin), Turn-Back Viewpoint (Zhongzhiyuan), Signal-Tower Honour Wall (Dazhongsil).
Character: tone—frontage—roof—public art; wayfinding uses the rail vocabulary; all brands/assets require clearance.

Renewal Projects, Implementation Policy, and Phasing

8 conceptual renewal projects: JZ-01..JZ-08 covering main-track stitching, yard frontages, departure platform, four-quadrant link, sandbox, switch posts, data counter, convention route.

Phasing: Stage 1 yard throats & main track; Stage 2 road/switch weaving; Stage 3 wing hinterlands.

Operations: annual convention, quarterly dispatch, monthly commissioning, weekly switch inspection; developer community via open timetable; all arrangements conceptual.

Metrics, Area Recalculation, and Compliance Matrix

Three classes: spatial recomputation, pending control metrics, operation performance metrics.

Known metrics recomputed from geometry under EPSG:4548; compliance matrix covers 23 mandatory tasks; standard matrix addresses all mandatory standards; design-depth matrix marks all 15 items complete.



Risk, Copyright, and Compliance

Main risks: full recalculation from provisional boundary; missing approved controls, redlines, ownership, utilities; data privacy & consent; heritage and brand clearance.

Compliance: public/cleared sources only; author responsible for AI-generated content; no non-public drawings or personal data; conceptual suggestions never masquerade as approvals.

Full sources, assumptions and copyright statements are in sources.json, assumptions.json and report/copyright_statement.md.

